Week #9: [Gluteal Region & Thigh]

(Hyperlinked = anatomy notes below)

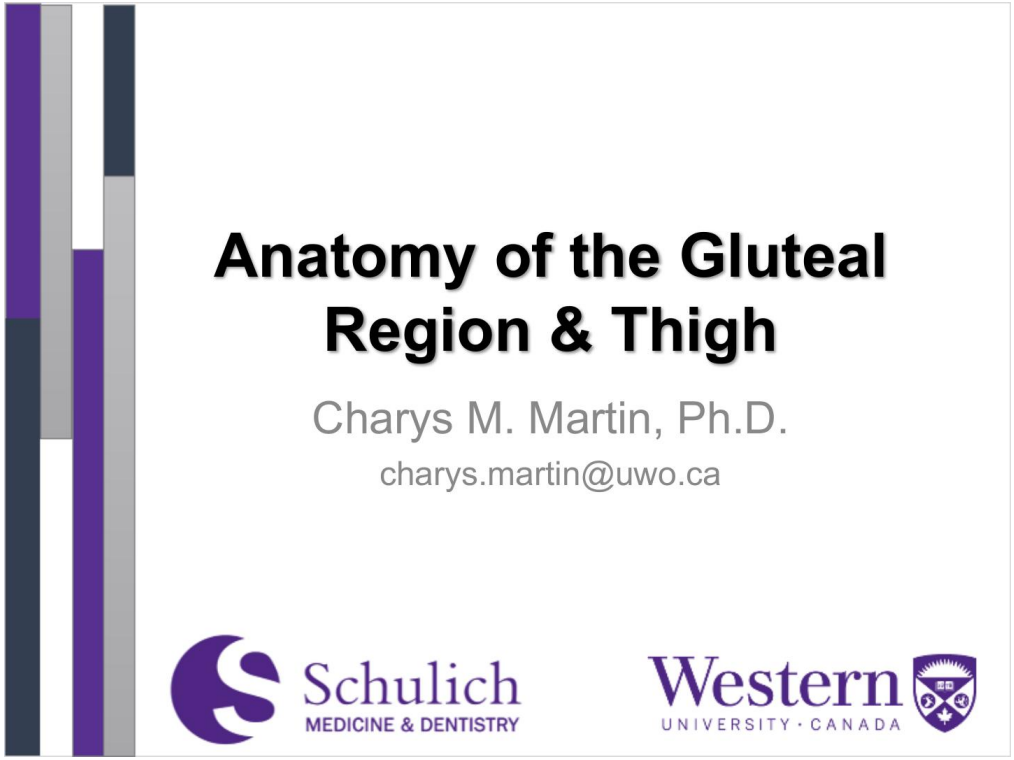
- Introduction to Module

Gluteal Region

- Review of Ligaments & Spaces of the Pelvis (7 min)
- Musculature (13 min)
- Neurovasculature (11 min)

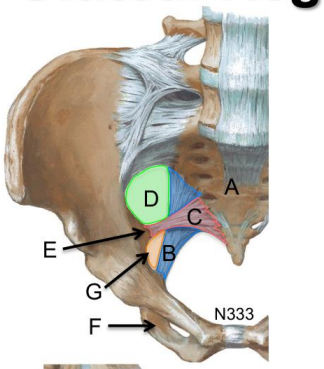
Thigh Region

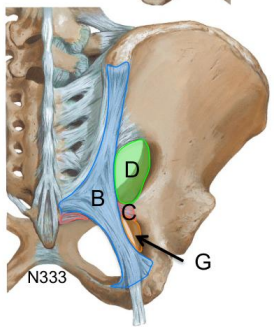
- Posterior Compartment (26 min)
- Anterior Compartment (19 min)
- Medial Compartment (13 min)
- Vasculature and Lymphatics of the Anteromedial Thigh (23 min)

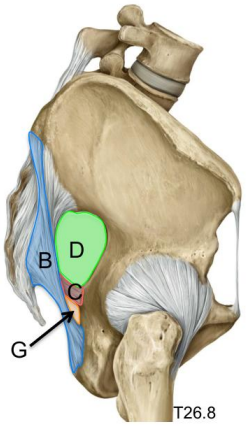


Review of Ligaments & Spaces of the Pelvis

Gluteal Region: Ligaments/Spaces







Ischial Spine

Ischial Tuberosity

Greater Sciatic Foramen

Lesser Sciatic Foramen

Sacrum

Sacrospinous Ligament

Sacrotuberous Ligament

I. Gluteal Region:

A. Gluteal Ligaments:

a. **Sacrotuberous ligament** – extends between the sacrum and ischial tuberosity. Converts the sciatic notches into greater and lesser sciatic foramina.

b. **Sacrospinous ligament** – Extends between the sacrum and ischial spine; extends deep to the sacrotuberous ligament. Converts the sciatic notches into greater and lesser sciatic foramina.

B. Anatomical Spaces:

1. **Greater Sciatic Foramen** – bounded anteriorly by the greater sciatic notch, posteriorly by the sacrotuberous ligament, and inferiorly by the sacrospinous ligament; provides a passageway for structures to enter the gluteal region from the pelvis.
2. **Lesser Sciatic Foramen –** bounded anteriorly by the lesser sciatic notch, superiorly by the ischial spine and sacrospinous ligament, and posteriorly by the sacrotuberous ligament; provides a passageway for structures to enter the perineum from the gluteal region.

B. Anatomical Spaces:

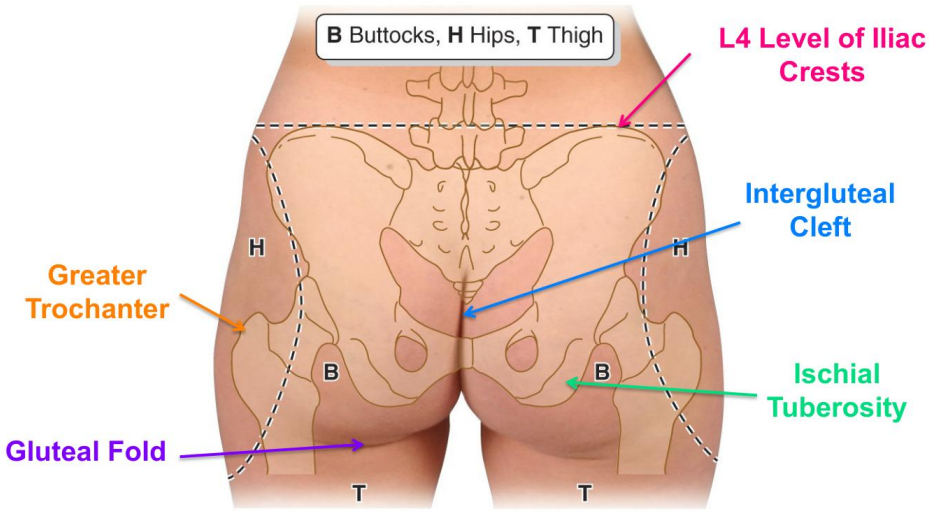
1. **Greater Sciatic Foramen** – bounded anteriorly by the greater sciatic notch, posteriorly by the sacrotuberous ligament, and inferiorly by the sacrospinous ligament; provides a passageway for structures to enter the gluteal region from the pelvis.
2. **Lesser Sciatic Foramen** – bounded anteriorly by the lesser sciatic notch, superiorly by the ischial spine and sacrospinous ligament, and posteriorly by the sacrotuberous ligament; provides a passageway for structures to enter the perineum from the gluteal region.
3. **Structures passing through the foramina:**

Greater Sciatic Foramen	**Lesser Sciatic Foramen**
Piriformis Superior & Inferior Gluteal vessels and nerves Sciatic Nerve Posterior Cutaneous Nerve of the Thigh Nerve to Quadratus Femoris Pudendal Nerve* Nerve to Obturator Internus* Internal Pudendal Vessels*	Pudendal Nerve* Nerve to Obturator Internus* Internal Pudendal Vessels*

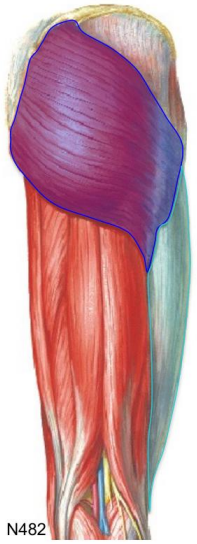
***** Passes through both the greater and lesser sciatic foramen

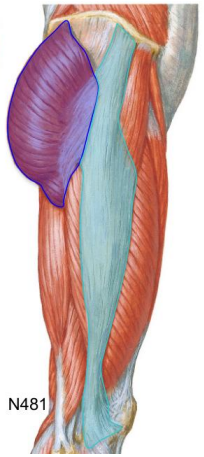
Musculature

Gluteal Region: Surface Anatomy



Gluteal Muscles: Gluteus Maximus

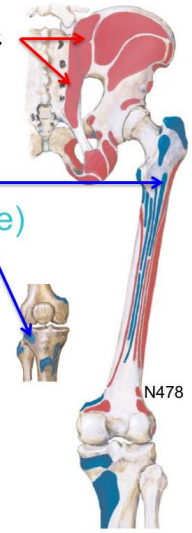




Myotome: S1

Origin:
Ilium, Sacrum, Coccyx, & Sacrotuberous Ligament

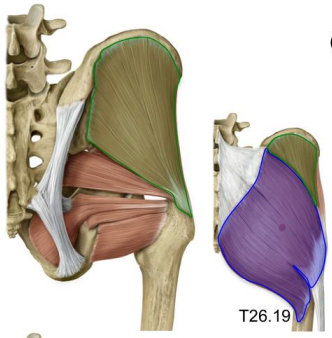
Insertion:
Gluteal Tuberosity
IT Band (Gerdy's Tubercle)

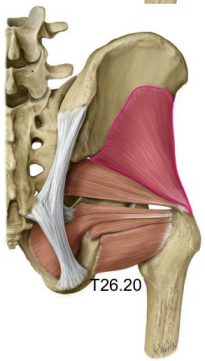


Innervation:
Inferior Gluteal Nerve

Function:
- Extension of hip
- Assists in standing from seated position

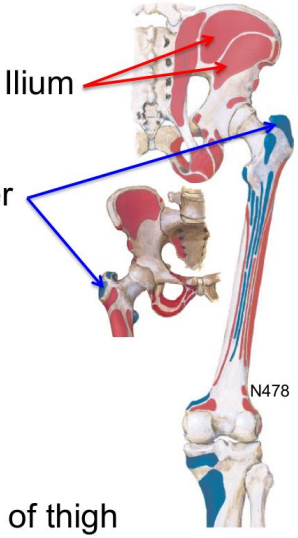
Gluteal Muscles: Gluteus Medius & Minimus





Origin:
Lateral Surface of Ilium

Insertion:
Greater Trochanter

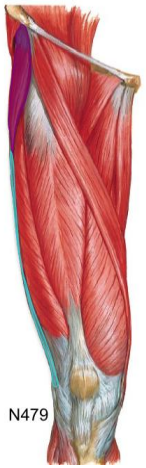


Innervation:
Superior Gluteal Nerve

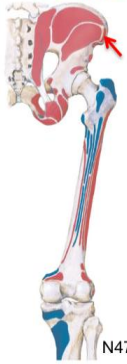
Function:
- Abduction & medial rotation of thigh
- Gait stabilization

Gluteal Muscles: Tensor Fascia Latae





Myotome: L5



Origin:
ASIS & Anterior Iliac Crest

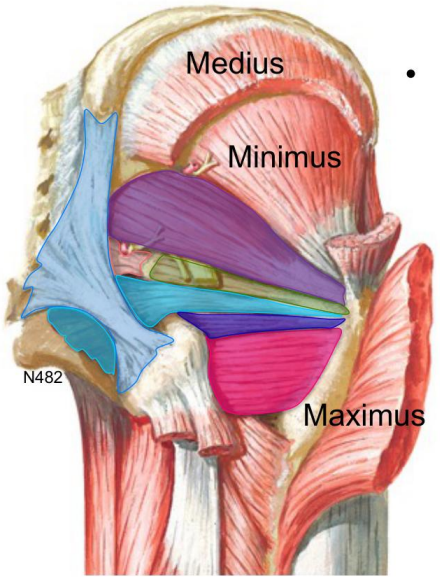
Insertion:
IT Band
(Gerdy's Tubercle)



Innervation: Superior Gluteal Nerve

Function:
- Flexes hip
- Tenses IT band

Gluteal Muscles: Lateral Rotators



- Group of muscles deep to Gluteus Maximus
- Function:
  - Laterally rotate hip
  - Stabilize the head of the femur in the acetabulum

Piriformis

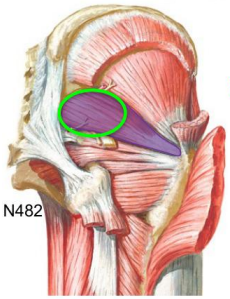
Superior Gemellus

Obturator Internus

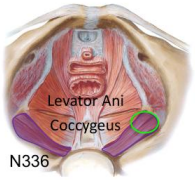
Inferior Gemellus

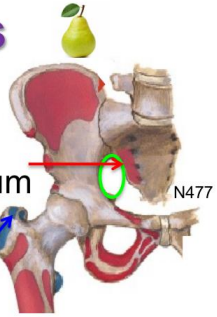
Quadratus Femoris

Gluteal Muscles: Piriformis



✧ Exists pelvis via
Greater Sciatic Foramen

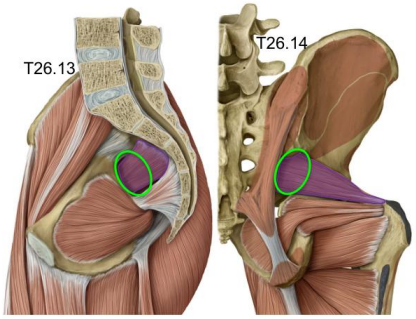




Origin:
Anterior Sacrum

Insertion:
Greater Trochanter

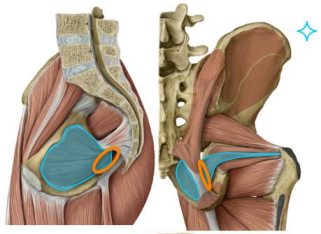
✧ **Important landmark in gluteal region**



Innervation: Nerve to Piriformis

Function:
- Lateral rotation of thigh
- Abduction of thigh
- Stabilization of hip

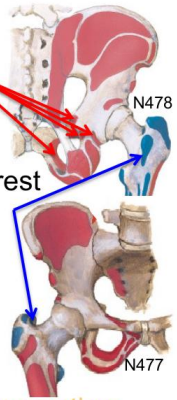
Gluteal Muscles: Lateral Rotators



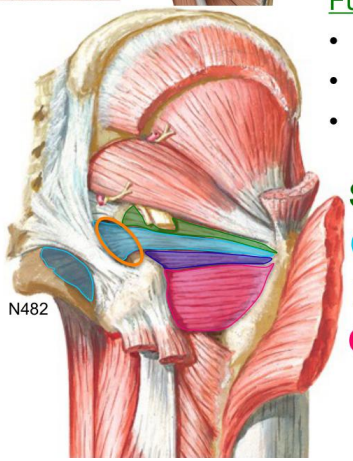
✧ **OI** - Exists pelvis via
Lesser Sciatic Foramen

Origin:
Ischium

Insertion:
Intertrochanteric crest
Trochanteric fossa



Function:
- Lateral rotation of thigh
- Abduction of thigh
- Stabilization of hip



Superior Gemellus } Nerve to Obturator Internus
Obturator Internus }

Inferior Gemellus } Nerve to Quadratus Femoris
Quadratus Femoris }

Innervation:

Knowledge Check:

A 72-year-old woman presents to the clinic two weeks after undergoing surgery for a total hip replacement. She is complaining of a 'limp' while walking. You ask her to walk for you. While walking you notice that her right hip drops every time her right leg leaves the ground for the swing phase. Which of the following muscles is not functioning properly? Trendelenburg Sign/Trendelenburg Gait

- What do you see?
  - Hips do not stay lvl during swing phase
- What do you think could be happening anatomically?
  - weakness in muscles that stabilize the pelvis during gait
- Which muscle(s) are likely involved?
  - gluteus medius + minimus
- What side is the weakness on?
  - Left side
- What is the innervation of these muscles?
  - Superior gluteal nerve



Muscles:

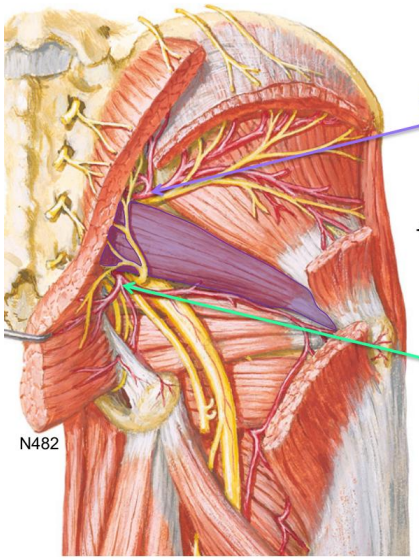
Muscle	Origin	Insertion	Innervation	Function
Gluteus Maximus	Ilium, Sacrum, coccyx, & sacrotuberous ligament	Lateral tibial condyle via IT band & gluteal tuberosity	Inferior Gluteal Nerve	Extension of hip; assists in standing from a seated position
Gluteus Medius	Lateral surface of ilium	Greater trochanter	Superior Gluteal Nerve	**Ab**duction & medial rotation of hip; gait stabilization
Gluteus Minimus				
Tensor Fascia Latae	ASIS & anterior iliac crest	Lateral tibial condyle via IT Band		Flexes hip; tenses IT band
Piriformis	Anterior sacrum; passes through greater sciatic notch	Greater trochanter	Nerve to Piriformis	

E. Nervous Structures:

1. **Superior Gluteal Nerve** (L4-S1) - Exits the pelvis via the ***greater sciatic foramen*** <u>superior</u> to the ***piriformis*** and courses between the gluteus medius and minimus; innervates the gluteus medius, gluteus minimus, and tensor fasciae latae.
   - ❖ ***<u>Clinical Note:</u>*** Injury to the superior gluteal nerve, for example during hip replacement surgery, results in a disabling **Trendelenburg gait** as a result of weakened abduction of the thigh by the gluteus medius & minimus. When a person is asked to stand on one leg, the gluteus medius & minimus normally contract as soon as the contralateral foot leaves the floor to prevent tipping of the pelvis to the unsupported side. When a person with a lesion of the superior gluteal nerve is asked to stand on one leg, the pelvis descends on the unsupported side, indicating that the gluteus medius on the contralateral side is weak or nonfunctional. This is referred to clinically as a positive Trendelenburg Test. When the pelvis descends on the unsupported side, the lower limb becomes, in effect, too long and does not clear the ground when the foot is brought forward in the swing phase of walking. To compensate, the individual leans away from the unsupported side, raising the pelvis to allow adequate room for the foot to clear the ground as it swings forward; this is known clinically as a gluteus medius gait.
2. **Inferior Gluteal Nerve** (L5-S2) - Exits the pelvis via the ***greater sciatic foramen*** <u>inferior</u> to ***piriformis***; innervates gluteus maximus.
3. **Nerve to Obturator Internus** (L5-S2) - Exits the pelvis via the ***greater sciatic foramen*** <u>inferior</u> to ***piriformis***; innervates superior gemellus and obturator internus.
4. **Nerve to Quadratus Femoris** (L4-S1) - Exits the pelvis via the ***greater sciatic foramen*** <u>inferior</u> to the ***piriformis***; innervates inferior gemellus and quadratus femoris.
5. **Pudendal Nerve (S2-S4)** - Exits the pelvis via the ***greater sciatic foramen*** <u>inferior</u> to ***piriformis***, then enters the perineum via the ***lesser sciatic foramen***; runs with ***internal pudendal artery***; innervates the perineum.
6. **Sciatic Nerve (L4-S3) -** Exits the pelvis via the ***greater sciatic foramen*** <u>inferior</u> to ***piriformis***,

Neurovasculature

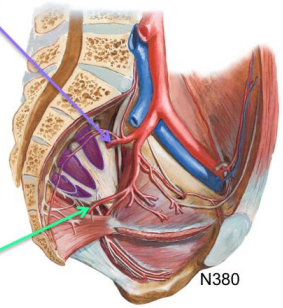
Gluteal Region: Arterial Supply



Superior Gluteal Artery

✧ Important landmark: Piriformis

Inferior Gluteal Artery

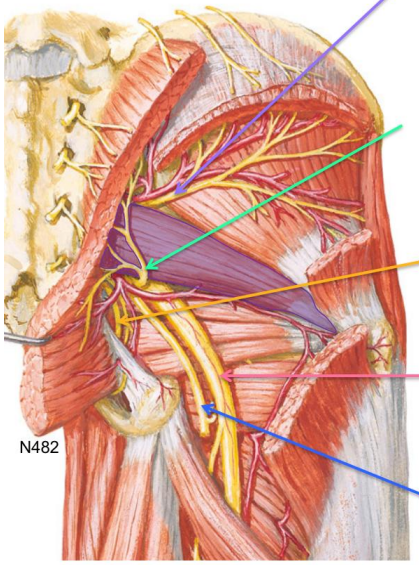


D. Vasculature:

1. **Superior Gluteal Artery –** branches from the internal iliac artery, exits the pelvis via the ***greater sciatic foramen*** superior to ***piriformis***; supplies the gluteal region.
2. **Inferior Gluteal Artery** – branches from the internal iliac artery; exits the pelvis via the ***greater sciatic foramen*** inferior to ***piriformis***; supplies the gluteal region.
3. **Internal Pudendal Artery** – branches from the internal iliac artery, exits the pelvis via the ***greater sciatic foramen*** inferior to ***piriformis***, then enters the perineum via the ***lesser sciatic foramen***; supplies the perineum.

Gluteal Region: Nervous Structures

✧ Important landmark: Piriformis



Superior to Piriformis

Superior Gluteal Nerve

Inferior to Piriformis

Inferior Gluteal Nerve

Pudendal Nerve (S2-S4)

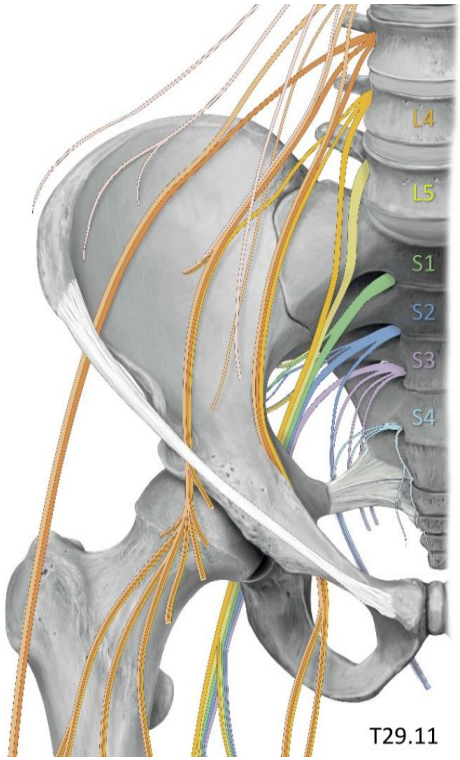
Sciatic Nerve (L4-S3)

Posterior Cutaneous Nerve of the Thigh



Sciatic Nerve

- Formed from anterior rami of spinal nerves L4-S3
- Largest Nerve in the Body composed of two nerves:
  - Tibial Nerve
  - Common Peroneal Nerves
- Sciatic nerve injury often results in pain or numbness in the buttock that radiates down the leg.
- Causes:
  - Herniated Disc
  - Hip Surgery
  - Piriformis Syndrome
  - Intragluteal Injection



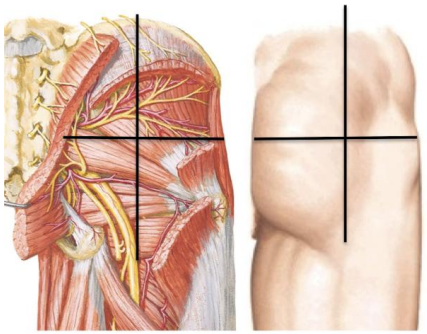
E. Nervous Structures:

1. **Superior Gluteal Nerve** (L4-S1) - Exits the pelvis via the ***greater sciatic foramen*** superior to the ***piriformis*** and courses between the gluteus medius and minimus; innervates the gluteus medius, gluteus minimus, and tensor fasciae latae.

Clinical Note: Injury to the superior gluteal nerve, for example during hip replacement surgery, results in a disabling **Trendelenburg gait** as a result of weakened abduction of the thigh by the gluteus medius & minimus. When a person is asked to stand on one leg, the gluteus medius & minimus normally contract as soon as the contralateral foot leaves the floor to prevent tipping of the pelvis to the unsupported side. When a person with a lesion of the superior gluteal nerve is asked to stand on one leg, the pelvis descends on the unsupported side, indicating that the gluteus medius on the contralateral side is weak or nonfunctional. This is referred to clinically as a positive Trendelenburg Test. When the pelvis descends on the unsupported side, the lower limb becomes, in effect, too long and does not clear the ground when the foot is brought forward in the swing phase of walking. To compensate, the individual leans away from the unsupported side, raising the pelvis to allow adequate room for the foot to clear the ground as it swings forward; this is known clinically as a gluteus medius gait.

2. **Inferior Gluteal Nerve** (L5-S2) - Exits the pelvis via the ***greater sciatic foramen*** inferior to ***piriformis***; innervates gluteus maximus.
3. **Nerve to Obturator Internus** (L5-S2) - Exits the pelvis via the ***greater sciatic foramen*** inferior to ***piriformis***; innervates superior gemellus and obturator internus.
4. **Nerve to Quadratus Femoris** (L4-S1) - Exits the pelvis via the ***greater sciatic foramen*** inferior to the ***piriformis***; innervates inferior gemellus and quadratus femoris.
5. **Pudendal Nerve (S2-S4)** - Exits the pelvis via the ***greater sciatic foramen*** inferior to ***piriformis***, then enters the perineum via the ***lesser sciatic foramen***; runs with ***internal pudendal artery***; innervates the perineum.
6. **Sciatic Nerve (L4-S3)** - Exits the pelvis via the ***greater sciatic foramen*** inferior to ***piriformis***, and descends into the posterior thigh. Largest nerve in the body and actually composed of two nerves within a common sheath:
    i. **Tibial nerve**
    ii. **Common Peroneal (Fibular) Nerve**
    b. ***Clinical Note:*** Injury to the sciatic nerve (sciatica) may cause pain, tingling or numbness from the buttock to the foot on the injured side. One cause may be **piriformis syndrome**, which results when the piriformis muscles compresses the nerve as it exits the pelvis. The sciatic nerve usually exits below piriformis; however, the nerve may pass completely or partially through piriformis where it can become compressed. Piriformis syndrome is common in athletes that require excessive use of the gluteal muscles. Treatments include NSAIDs, rest, and proper stretching.
7. **Posterior Cutaneous Nerve of the Thigh** (S1-S3) - Exits the pelvis via the ***greater sciatic foramen*** inferior to the ***piriformis*** and travels deep to gluteus maximus; innervates the skin of the inferior buttocks, posterior thigh and popliteal fossa.
    a. ***Clinical Note:*** **Intragluteal injections:** the gluteal region is a common site for intramuscular injections because the muscles are thick and provide a large area for venous absorption. Injections are given in the superolateral quadrant of the buttock in order to avoid damage to the sciatic nerve. To find this quadrant, the index finger is placed on the ASIS and the fingers are spread posterolaterally along the iliac crest. The injection can then be made in the area between the index and middle finger.

Knowledge Check:



The gluteal region is a common site for intramuscular injections because the muscles are thick and provide a large area for venous absorption. In which of the following quadrants would you chose to provide an injection?

A. Lower Left Quadrant
B. Lower Right Quadrant
C. Upper Left Quadrant
D. Upper Right Quadrant

- to avoid damage to sciatic nerve

and descends into the posterior thigh. Largest nerve in the body and actually composed of two nerves within a common sheath:

a. **Tibial nerve**
b. **Common Peroneal (Fibular) Nerve**

❖ ***Clinical Note:*** Injury to the sciatic nerve (sciatica) may cause pain, tingling or numbness from the buttock to the foot on the injured side. One cause may be **piriformis syndrome**, which results when the piriformis muscles compresses the nerve as it exits the pelvis. The sciatic nerve usually exits below piriformis; however, the nerve may pass completely or partially through piriformis where it can become compressed. Piriformis syndrome is common in athletes that require excessive use of the gluteal muscles. Treatments include NSAIDs, rest, and proper stretching.

7. **Posterior Cutaneous Nerve of the Thigh** (S1-S3) - Exits the pelvis via the ***greater sciatic foramen*** inferior to the ***piriformis*** and travels deep to gluteus maximus; innervates the skin of the inferior buttocks, posterior thigh and popliteal fossa.

❖ ***Clinical Note:*** **Intragluteal injections:** the gluteal region is a common site for intramuscular injections because the muscles are thick and provide a large area for venous absorption. Injections are given in the superolateral quadrant of the buttock in order to avoid damage to the sciatic nerve. To find this quadrant, the index finger is placed on the ASIS and the fingers are spread posterolaterally along the iliac crest. The injection can then be made in the area between the index and middle finger.

II. Posterior Compartment of the Thigh:

A. Thigh Organization: The deep structures of the thigh are surrounded by deep strong fascia called **fascia lata**. Three extensions of this fascia (medial, lateral, and posterior septa) extend between the muscles to form three muscular compartments:

1. **Anterior Compartment** – lies anterior to the medial and lateral intermuscular septa. As a group, muscles in this compartment function to flex the hip and extend the knee and they are innervated by femoral nerve.
2. **Medial Compartment** – lies medial to the medial and posterior intermuscular septa. As a group, muscles in this compartment function to a**d**duct the hip and they are innervated by obturator nerve (with two exceptions: hamstring portion of adductor magnus and pectineus).
3. **Posterior Compartment** – lies posterior to the posterior and lateral intermuscular septa. As a group, muscles in this compartment function to **extend the hip** and **flex the knee** and are innervated by branches of the **sciatic nerve**.

B. Posterior Compartment:

1. **Muscles:**

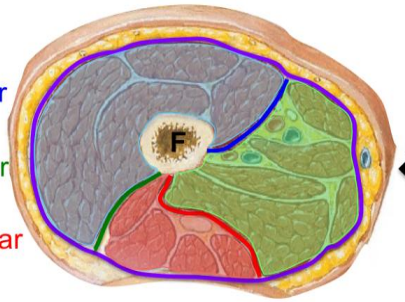
Posterior Compartment

- Muscles
- Neurovasculature

Thigh Organization

Anterior Compartment (AC):
Function: Flex hip & extend knee
Innervation: Femoral Nerve

Medial Compartment (MC):
Function: Adduct hip
Innervation: Obturator Nerve

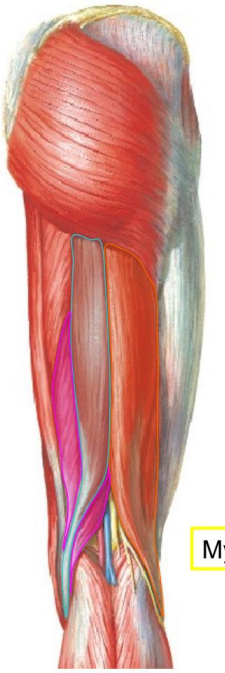


Fascia Lata
Medial Intermuscular Septum
Lateral Intermuscular Septum
Posterior Intermuscular Septum



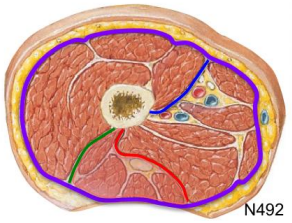
Posterior Compartment (PC):
Function: Extend hip & Flex Knee
Innervation: Sciatic Nerve

Posterior Compartment (PC): Muscles



- Hamstring Muscle Group
- Three Muscles
- Primarily responsible for **extension of the hip** and **flexion of the knee**
- Innervated by: Branches of Sciatic Nerve

Myotome: L5



II. Posterior Compartment of the Thigh:

A. Thigh Organization: The deep structures of the thigh are surrounded by deep strong fascia called **fascia lata**. Three extensions of this fascia (medial, lateral, and posterior septa) extend between the muscles to form three muscular compartments:

1. **Anterior Compartment –** lies anterior to the medial and lateral intermuscular septa. As a group, muscles in this compartment function to flex the hip and extend the knee and they are innervated by femoral nerve.
2. **Medial Compartment –** lies medial to the medial and posterior intermuscular septa. As a group, muscles in this compartment function to adduct the hip and they are innervated by obturator nerve (with two exceptions: hamstring portion of adductor magnus and pectineus).
3. **Posterior Compartment –** lies posterior to the posterior and lateral intermuscular septa. As a group, muscles in this compartment function to **extend the hip** and **flex the knee** and are innervated by branches of the **sciatic nerve.**

PC: Biceps Femoris



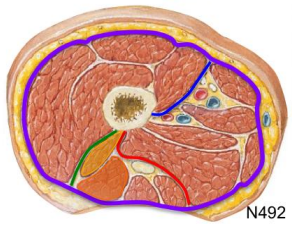


Origin:
LH: Ischial Tuberosity
SH: Linea Aspera

Insertion:
Head of Fibula

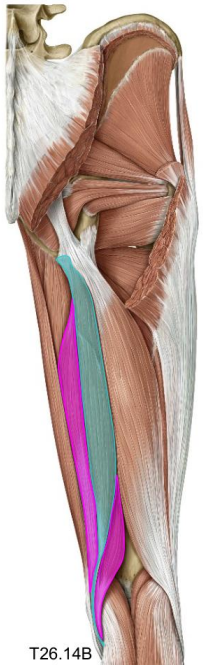


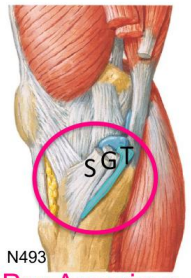
Innervation:
LH: Tibial division of Sciatic Nerve
SH: Common peroneal division of Sciatic Nerve



Function:
- Extension of hip
- Flexion of knee

PC: Semimembranosus & Semitendinosus

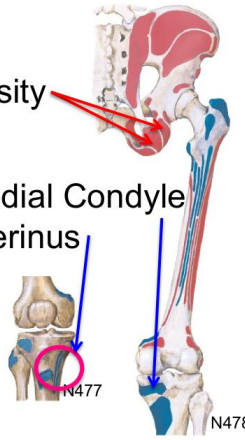




Pes Anserinus
- Sartorius
- Gracilis
- Semitendinosus

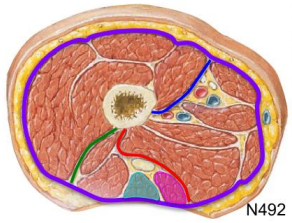
Origin:
Ischial Tuberosity

Insertion:
SM - Tibial Medial Condyle
ST – Pes Anserinus



Innervation:
Tibial division of Sciatic Nerve

Function:
- Extension of hip
- Flexion of knee

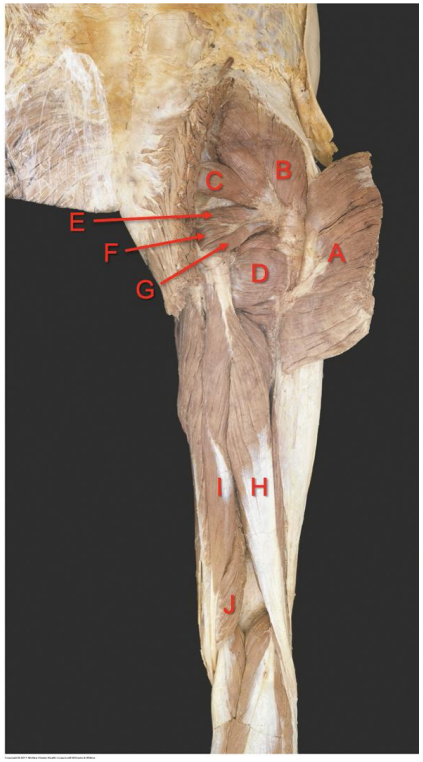


1. Muscles:

Muscle	Origin	Insertion	Innervation	Function
Biceps Femoris	Long Head: Ischial Tuberosity Short Head: Linea Aspera	Head of Fibula	Long Head: Tibial division of sciatic nerve Short Head: Common peroneal division of sciatic nerve	Extension of hip; flexion of knee
Semitendinosus	Ischial Tuberosity	Pes Anserinus	Tibial division of sciatic nerve	
Semimbranosus		Tibial Medial Condyle		

Knowledge Check:

Biceps Femoris (H)
Gluteus Maximus (A)
Gluteus Medius (B) fan shaped
Inferior Gemellus (G)
Obturator Internus (F) tendon end
Piriformis (C) pear shaped
Semimembranosus (J)
Semitendinosus (I) – mice tendon
Superior Gemellus (E)
Quadratus Femoris (D) – square-like straight from pelvis to trochanter



Knowledge Check



A 26 year old hurdler was practicing for an upcoming meet. As he was jumping a hurdle he feels a pop in the back of the thigh of his lead leg and he is unable to continue due to pain. He goes home to rest thinking it is a common hamstring strain.

Three days later he presents to the clinic with extensive bleeding over the posterior thigh. An x-ray shows an avulsion fracture indicated by the red arrow



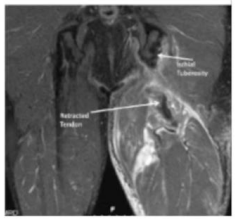
Hamstring Avulsion



Mechanism of Injury:
- Active contraction + passive stretch of hamstrings

Symptoms:
- Hematoma
- Pain

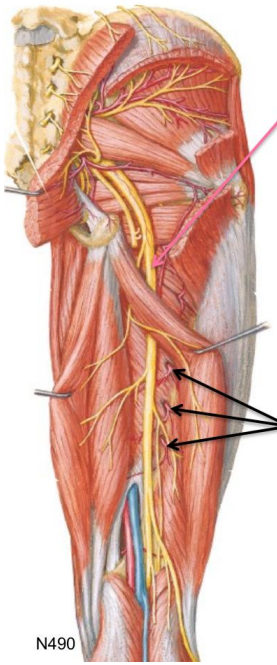
Treatment:
- Surgical reattachment



Note: **Pes Anserinus** (goose's foot) – refers to the conjoined tendons of three muscles (Sartorius, gracilis, and semitendinosus) that insert onto the anteromedial surface of the proximal tibia.

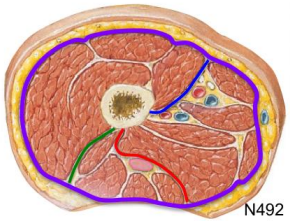
Clinical Note: **Hamstring Strains –** common injury in athletes. Muscle strains are a result of active contraction and passive stretch. Hamstring strains are associated with extreme pain and hematomas caused by the tearing of muscle fibres resulting in the rupture of blood vessels supplying the muscles. Forcible flexion of the hip with the knee extended may result in an avulsion fracture of the common hamstring origin the ischial tuberosity.

PC: Neurovasculature Structures



Sciatic Nerve
Comprised of:
- Tibial Division
  - Supplies SM, ST, Long head of BF
- Common Peroneal Division
  - Supplies Short head of BF

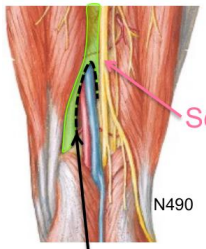
Perforating branches of
Profunda Femoris Artery



Nervous Structures:

a. **Sciatic Nerve (L4-S3)** – The two divisions of the sciatic nerve, ***tibial nerve*** and ***common peroneal nerve*** travel through the posterior compartment of the thigh in a common sheath deep to the long head of biceps femoris. The tibial nerve supplies all muscles in the posterior compartment except the short head of biceps femoris, which is supplied by the common peroneal nerve.

Popliteal Fossa

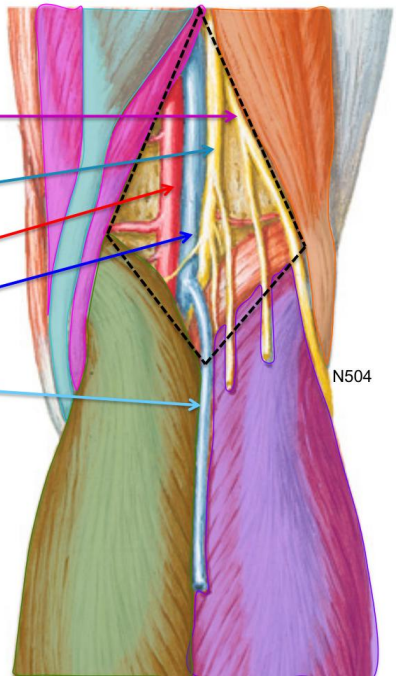


Sciatic Nerve

Adductor Hiatus via
Adductor Magnus

Relationships:
Deep → Superficial
AVN

Contents:
- Common Peroneal Nerve
- Tibial Nerve
- Popliteal Artery
- Popliteal Vein
- Small Saphenous Vein



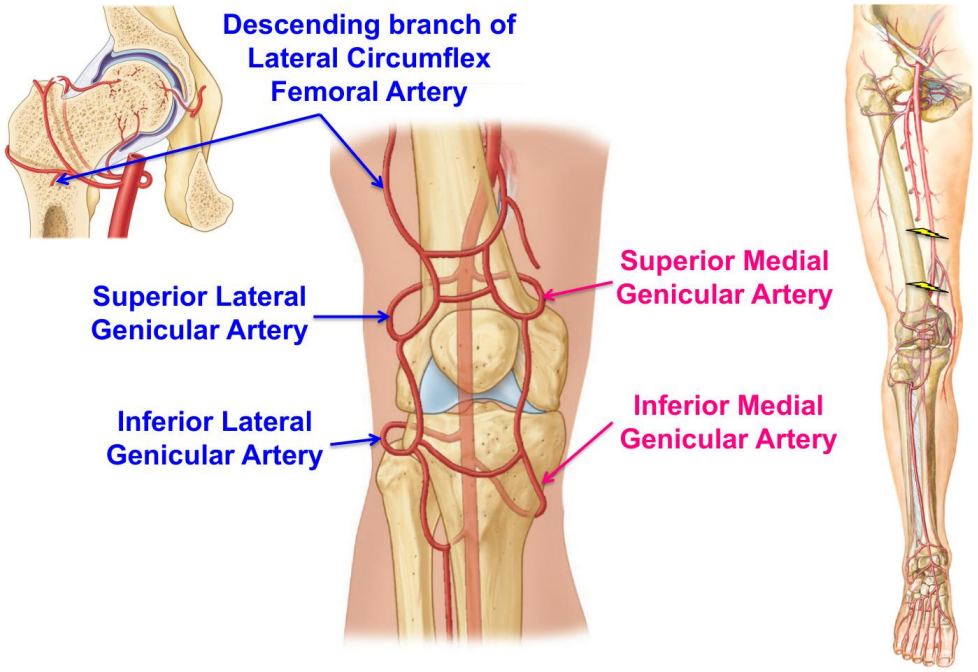
Boundaries:
- Biceps Femoris
- Semimembranosus & Semitendinosus
- Lateral Head of Gastrocnemius
- Medial Head of Gastrocnemius

Vasculature:

a. **Perforating Arteries** (branches of **Profunda Femoris Artery** (deep femoral artery)) – pierce adductor magnus to enter the posterior compartment.
   i. The posterior compartment of the thigh does not have isolated blood supply. It receives blood from the inferior gluteal, medial circumflex femoral, perforating, and popliteal arteries.

b. **Popliteal Artery and Vein** – continuation of the femoral artery and vein from the anterior compartment. These vessels pass through the adductor hiatus, which is an opening between the adductor magnus and the femur.

Popliteal Artery: Genicular Branches



Knowledge Check



An 85 year old presents at the clinic with a 6-week history of swelling of his right leg with no pain. Upon examination, he did have a palpable pulsatile mass in the upper right popliteal fossa of his right thigh. Which of the following is at the top of your differential diagnosis?

_____

- Most common _____
- At risk of _____, _____, and _____
- 80% asymptomatic
- _____

III. Popliteal Fossa

a. **Boundaries:**
   i. Superolaterally – Biceps Femoris
   ii. Superomedially – Semimembranosus and Semitendinosus
   iii. Inferiorly – Medial and Lateral Heads of the Gastrocnemius

b. **Contents:**
   i. Tibial and Common Peroneal Nerves
   ii. Popliteal Artery and Vein (Genicular branches)
   iii. Termination of the Small Saphenous Vein
   iv. Popliteal Lymph Nodes
   v. Fat

Clinical Note: The genicular anastomosis provides a network of vessels surrounding the knee that provides collateral circulation capable of maintaining blood supply to the leg during full knee flexion (which may temporarily occlude the popliteal artery) or if there is a pathological occlusion of the popliteal or femoral artery. The ***descending branch of the lateral femoral circumflex artery*** and the ***descending genicular artery*** (branch of the femoral artery) provide proximal contributions to the genicular anastomosis.

Clinical Note: A ***popliteal aneurysm*** is an abnormal dilation of all or part of the popliteal artery. It usually causes edema and pain in the popliteal fossa. Rupture of the artery is called a hemorrhage and if it is not treated quickly it can lead to loss of the foot and leg.

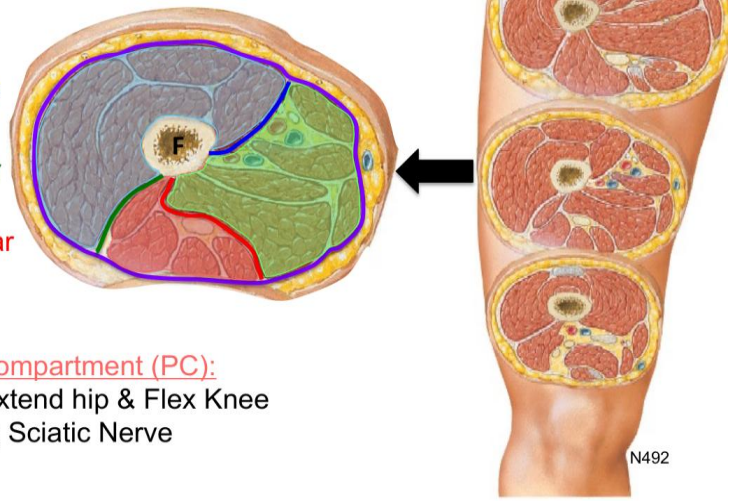
Anterior Compartment

- Muscles
- Neurovasculature

Thigh Organization

Anterior Compartment (AC):
Function: Flex hip & extend knee
Innervation: Femoral Nerve

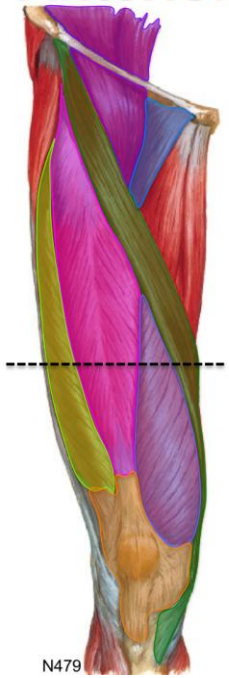
Medial Compartment (MC):
Function: Adduct hip
Innervation: Obturator Nerve



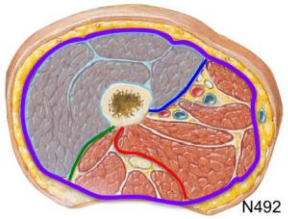
Fascia Lata
Medial Intermuscular Septum
Lateral Intermuscular Septum
Posterior Intermuscular Septum

Posterior Compartment (PC):
Function: Extend hip & Flex Knee
Innervation: Sciatic Nerve

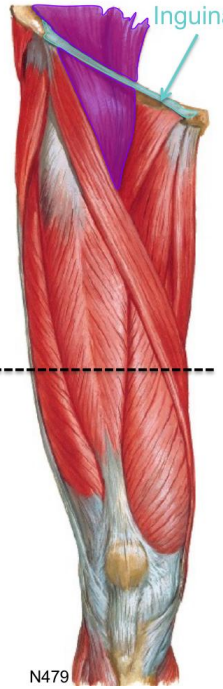
Anterior Compartment (AC): Muscles



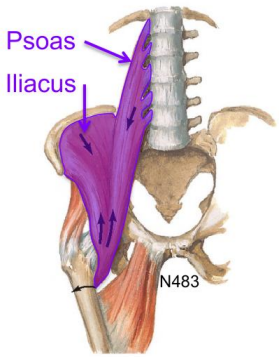
- Primarily responsible for **flexion of the hip** and **extension of the knee**
- Innervated by: **Femoral Nerve**



AC: Iliopsoas



Myotome: L1/2



Origin:
Psoas: T12-L5
Iliacus: Ilium

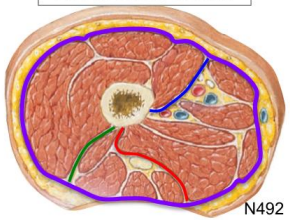
Insertion:
Lesser trochanter
via common tendon



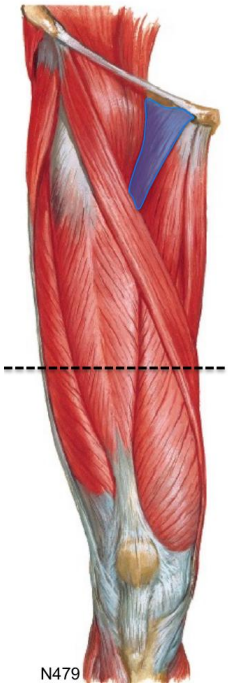
Innervation:
- Femoral Nerve
- Anterior Rami L1-L3

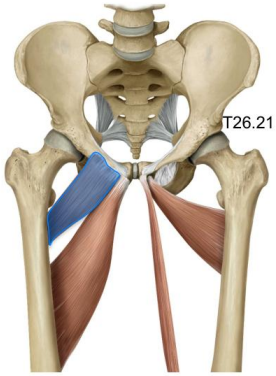
Function:
- Flexion of hip

Not in Cross-section

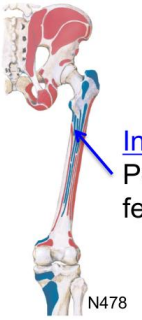


AC: Pectineus





Origin:
Pubic superior ramus



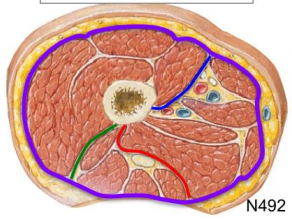
Insertion:
Posterior proximal
femur



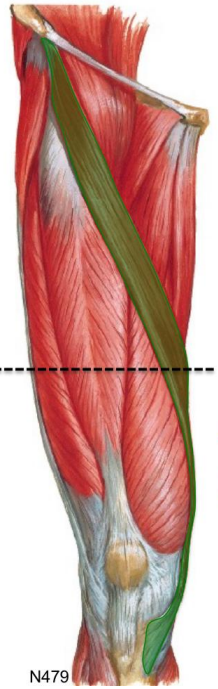
Innervation:
- Femoral Nerve

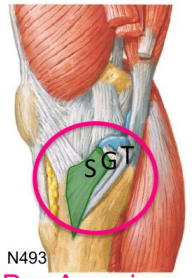
Function:
- Adduction of hip
- Flexion of hip

Not in Cross-section



AC: Sartorius





Pes Anserinus
- Sartorius
- Gracilis
- Semitendinosus

Origin: ASIS



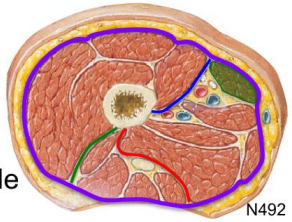
Insertion:
Pes Anserinus

Innervation: Femoral Nerve

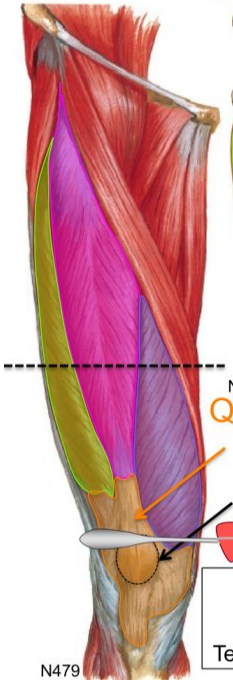
Function:
- Lateral rotation of hip
- Flexion of hip
- Flexion of knee
- Hip Abduction



"Tailor's" Muscle



AC: Quadriceps Femoris

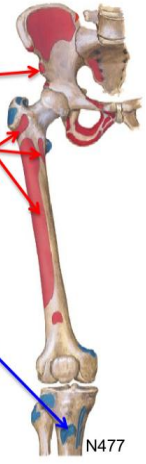


Myotome: L3/4

Rectus Femoris
Vastus Lateralis
Vastus Medialis
Vastus Intermedius

Patellar Tendon Reflex
Tests integrity of L3-L4

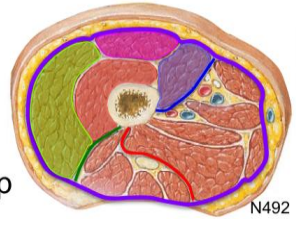
Origin:
RF: AIIS
Vastus Muscles:
Proximal Femur



Insertion:
Tibial tuberosity
via quadriceps
tendon

Innervation:
- Femoral Nerve

Function:
- Powerful knee extensor
- RF: Flex the hip



IV. Anterior Compartment

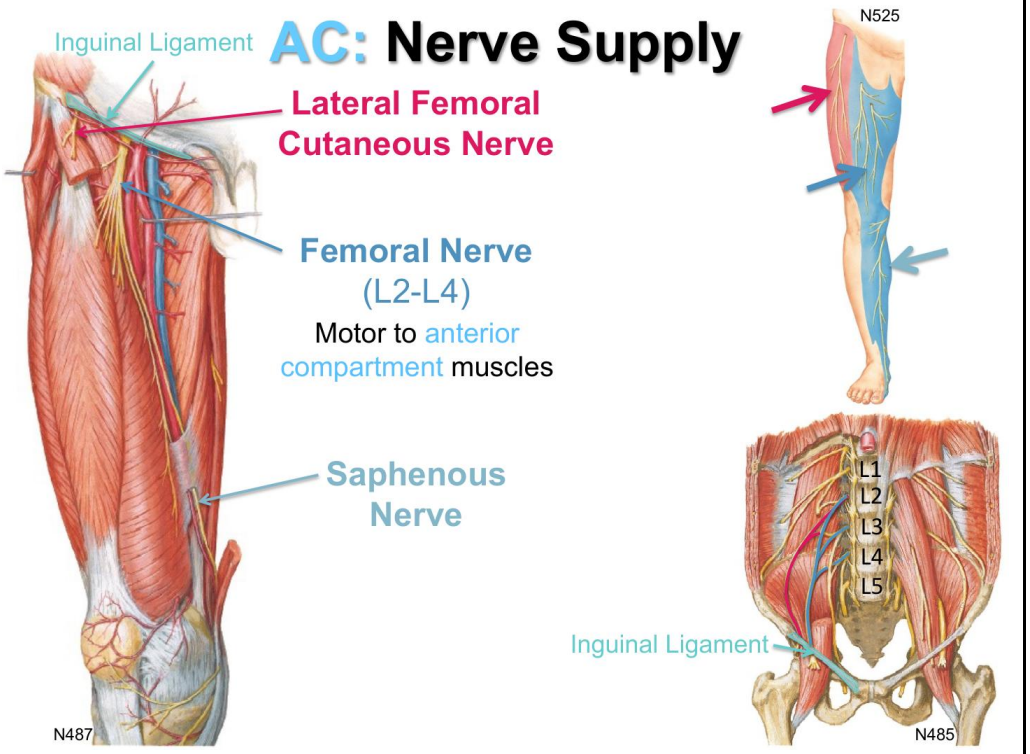
A. Muscles:

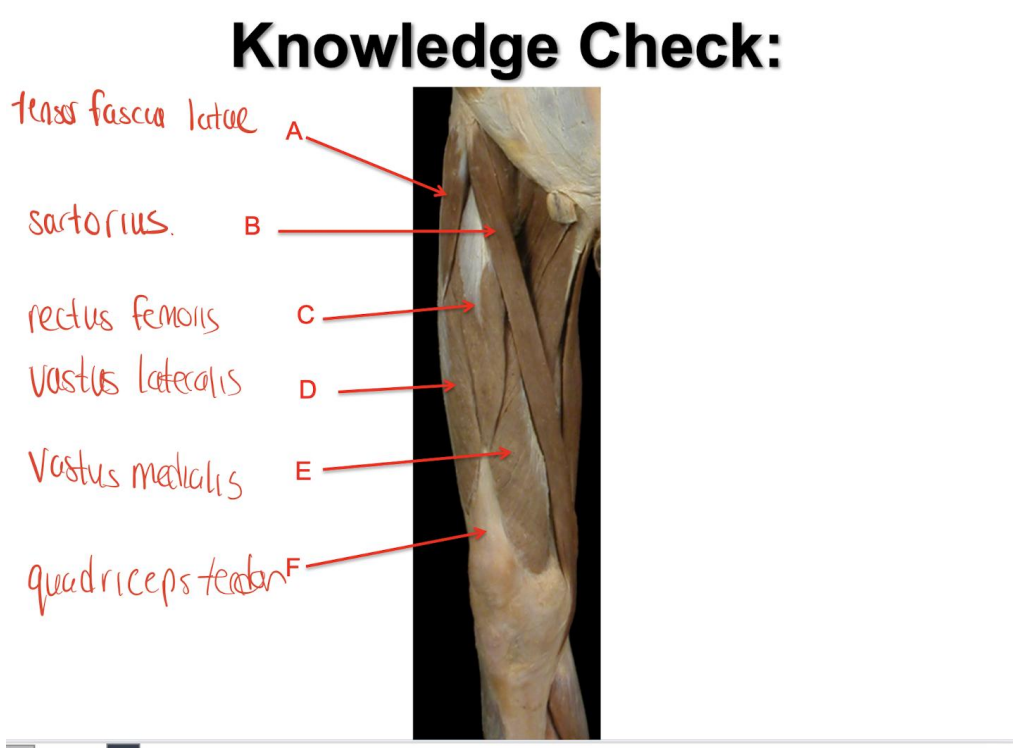
Muscle	Origin	Insertion	Action	Innervation
Pectineus	Superior ramus of pubis	Pectineal line of femur (just inferior to lesser trochanter)	Adducts and flexes thigh	Femoral Nerve
Sartorius	ASIS	Pes Anserinus	Flexes, abducts, and laterally rotates hip; flexes knee	Femoral Nerve
Iliopsoas	T12-L5 vertebrae and Ilium	Lesser trochanter of femur	Flexes the thigh at hip joint	Femoral Nerve
Quadriceps Femoris				
Rectus Femoris	AIIS	Tibial tuberosity via quadriceps tendon	Extends leg at knee joint (although compromised during hip flexion); helps iliopsoas flex thigh, stabilizes hip joint	Femoral Nerve
Vastus Lateralis	Greater trochanter and lateral lip of linea aspera	Tibial tuberosity via quadriceps tendon	Extends leg at knee joint	Femoral Nerve
Vastus Medialis	Intertrochanteric line and medial lip of linea aspera	Tibial tuberosity via quadriceps tendon	Extends leg at knee joint	Femoral Nerve
Vastus Intermedius	Shaft of femur	Tibial tuberosity via quadriceps tendon	Extends leg at knee joint	Femoral Nerve

- The **quadriceps (patellar) tendon** is the common tendon for all of the quadriceps muscles, which inserts on the tibial tuberosity. The central part of the quadriceps

tendon that extends from the patella to the tibial tuberosity may be called the patellar ligament.

- **Pes anserinus** (Latin – goose's foot) is the insertion of ***Sartorius, Gracilis,*** and ***Semitendinosus*** on the medial aspect of the tibia, superficial to the Medial Collateral Ligament of the knee.





B. Nervous Structures:

1. **Lateral Femoral Cutaneous Nerve (L2-L3):** Passes under inguinal ligament adjacent to anterior superior iliac spine to supply skin to lateral thigh.
2. **Femoral Nerve (L2-L4):** Innervates the anterior thigh muscles; originates from the lumbar plexus and passes deep to the inguinal ligament to enter the femoral triangle, lateral to femoral vessels. After passing the inguinal ligament it divides into:
    a. Muscular branches to anterior thigh
    b. Cutaneous branches to anterior thigh
    c. Saphenous Nerve – terminal cutaneous branch that descends through the femoral triangle and adductor canal. Supplies the skin and fascia on the anteromedial aspect of the knee, leg, and foot.

Clinical Note: The **patellar tendon reflex** is routinely tested during physical examinations by firmly striking the patellar ligament while the patient is sitting down with their legs dangling. A normal reflex would cause the quadriceps to contract and the leg to extend. This tendon reflex tests the integrity of the **femoral nerve** and the **L3-L4 spinal cord segments.** A diminution or absence of the patellar tendon reflex may indicate a lesion that interrupts the innervation of the quadriceps.

Medial Compartment

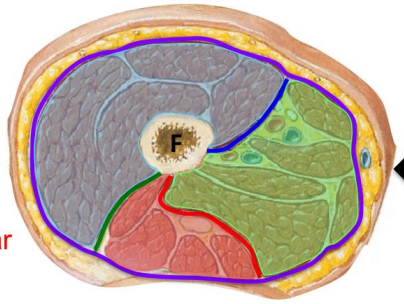
- Muscles
- Neurovasculature

Thigh Organization

Anterior Compartment (AC):
Function: Flex hip & extend knee
Innervation: Femoral Nerve

Medial Compartment (MC):
Function: Adduct hip
Innervation: Obturator Nerve

Fascia Lata
Medial Intermuscular Septum
Lateral Intermuscular Septum
Posterior Intermuscular Septum

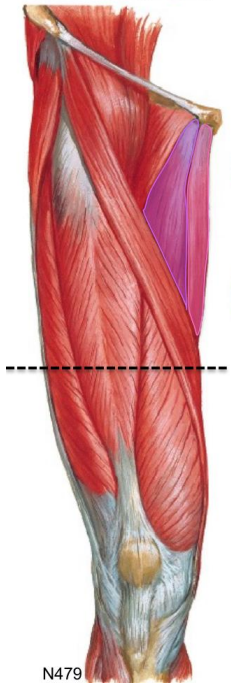


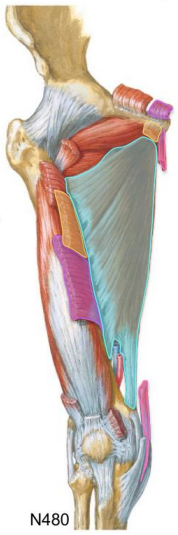


Posterior Compartment (PC):
Function: Extend hip & Flex Knee
Innervation: Sciatic Nerve

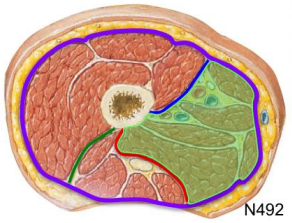
Medial Compartment (MC): Muscles

Myotome: L3

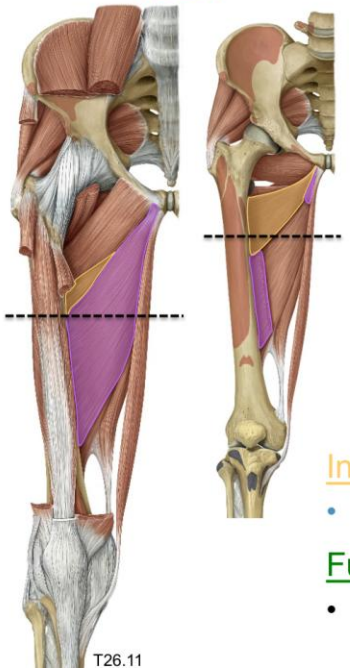




- Primarily responsible for **adduction of the hip**
- Innervated by: Obturator Nerve

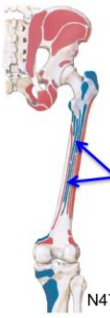


MC: Adductor Longus & Brevis



Origin:
Pubic Bone

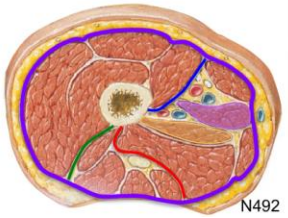
Insertion:
Linea Aspera



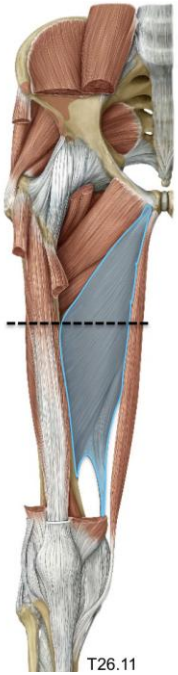


Innervation:
- Obturator Nerve

Function:
- Adduction of hip



MC: Adductor Magnus





Adductor Part
Hamstring Part
Adductor Hiatus

Origin:
AP: Ischium
HP: Ischial Tuberosity

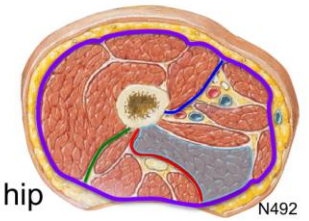


Insertion:
AP: Linea Aspera
HP: Adductor Tubercle

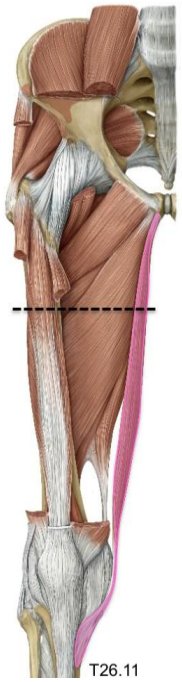


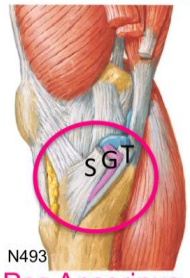
Innervation:
- AP: Obturator Nerve
- HP: Tibial Nerve

Function:
- AP: **Ad**duction and *flexion* of hip
- HP: **Ad**duction and *extension* of hip



MC: Gracilis





Pes Anserinus

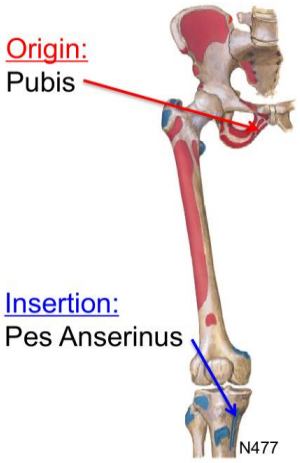
- Sartorius
- Gracilis
- Semitendinosus

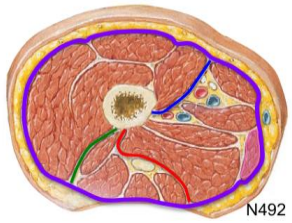
Innervation:

- Obturator Nerve

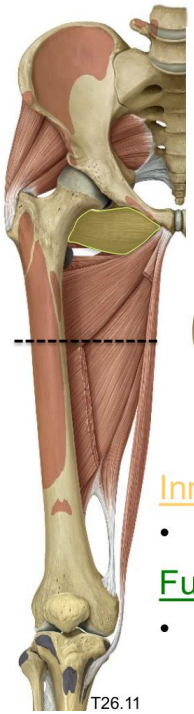
Function:

- Adduction of hip
- Flexion of knee
- Assists with medial rotation





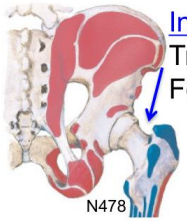
MC: Obturator Externus





Origin:
Obturator Foramen

Insertion:
Trochanteric Fossa



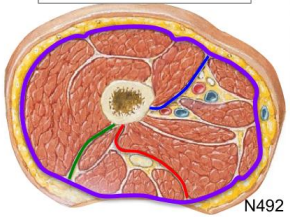


Innervation:
- Obturator Nerve

Function:
- Laterally rotates hip



Not in Cross-section



Knowledge Check:

Inserts on lesser trochanter



A iliopsoas

B pectineus

C adductor longus

D gracilis

Muscle	Origin	Insertion	Action	Innervation
Adductor Longus	Body of pubic bone	Linea Aspera	Adducts thigh	Obturator Nerve
Adductor Brevis	Body & inferior ramus of pubis	Pectineal line & linea aspera	Adducts thigh	Obturator Nerve
Adductor Magnus	**Adductor part:** inferior pubic ramus, ramus of ischium **Hamstring part:** ischial tuberosity	**Adductor part:** gluteal tuberosity, linea aspera, supracondylar line **Hamstring part:** adductor tubercle of femur	**Adductor part:** adducts and flexes the thigh **Hamstring part:** adducts and extends the thigh	**Adductor part:** Obturator Nerve **Hamstring part:** Tibial Nerve
Gracilis	Body and inferior pubic ramus	Superior part of medial surface of tibia via pes anserinus	Adducts thigh, flexes leg, and helps rotate the leg medially	Obturator Nerve
Obturator Externus	Margins of obturator foramen	Trochanteric fossa of femur	Laterally rotates thigh	Obturator Nerve

Groin Pull







- Active contraction and passive stretch of the medial compartment muscles
- Symptoms:
  - Bruising
  - Pain
- Treatment:
  - Conservative
    - RICE
    - NSAIDS

LO #7



Case #1

A 14 year old girl presents to the clinic complaining of pain in her right thigh and anterior pelvic region. She was kicking a football when severe pain in her right thigh rendered her unable to walk.

Upon examination there was tenderness over the anterior superior iliac spine.



ASIS Avulsion

muscles attach here

- Musculotendinous origin of sartorius and tensor fascia latae
- Most common in adolesce the growing apophysis
- Treatment:
  - Conservative or surgical fracture displacement

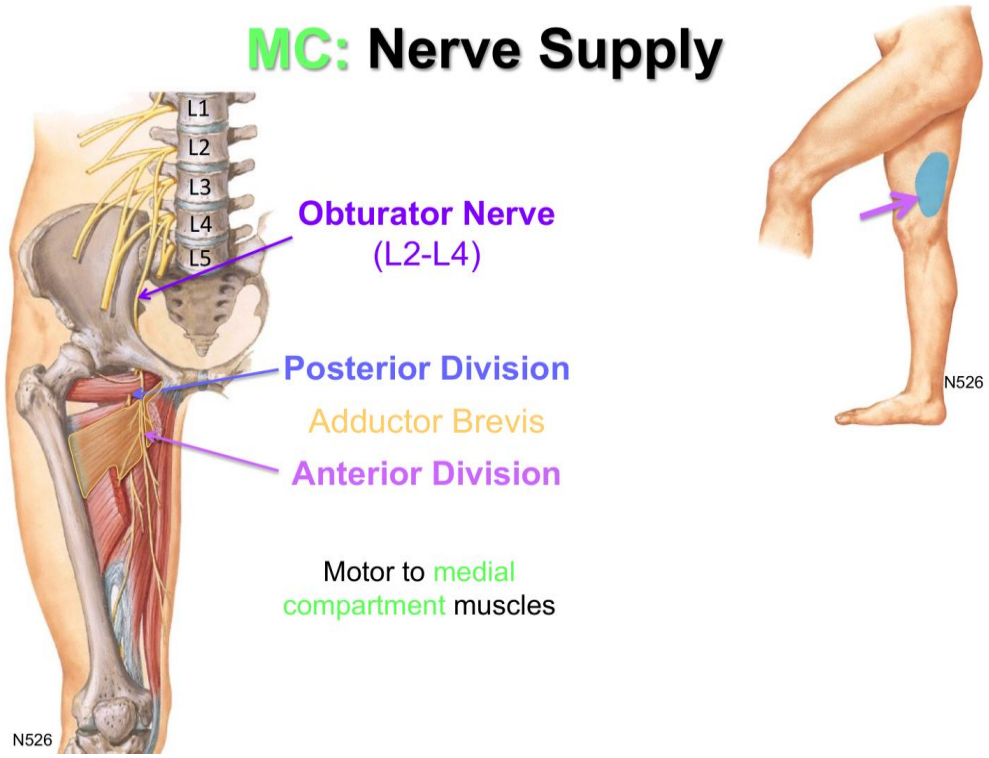
because still have



Clinical Note: **Gracilis** is a weak adductor and thus may be removed without a functional deficit. As a result gracilis can be transplanted, with its nerve and blood vessels, to replace a damaged muscle.

Clinical Note: A **groin pull** results from the tearing or stretching of the adductor muscles close to their origin. This injury is common in sports that require quick starts or extreme stretching, such as soccer, sprinting, and gymnastics.

Clinical Note: In teenagers the cartilaginous growth plate in bones is weaker than the musculotendinous units that are attached to the apophyses (growth centre) and thus powerful movements may fracture the bone. Strong powerful flexion of the hip may results in an avulsion of the ASIS, which is the origin of Sartorius and Tensor Fascia Latae muscles.

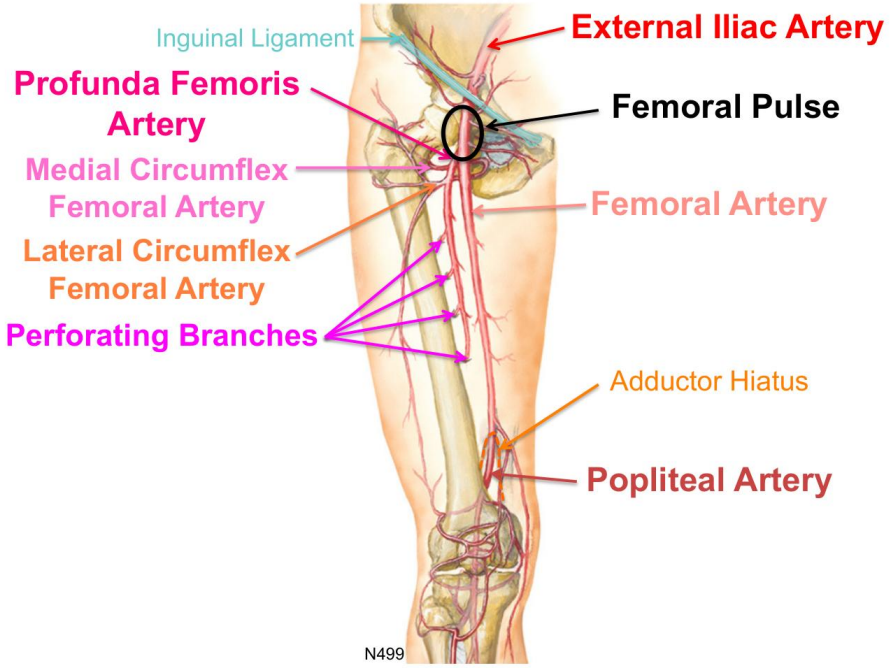


B. Nervous Structures:

1. Obturator Nerve (L2-L4): Descends medial to psoas major and passes through the obturator foramen (with obturator vessels) to enter the thigh; innervates the medial thigh muscles (except the hamstring portion of adductor magnus) and provides some sensory innervation to the medial thigh. Divides into **anterior** and **posterior** divisions, which straddle ***adductor brevis***.

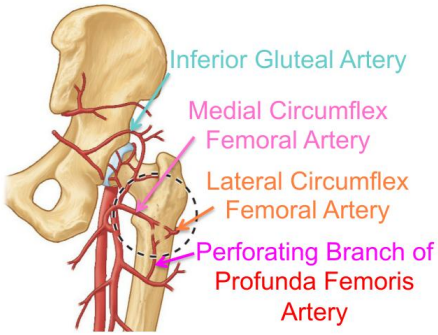
Vasculature and Lymphatics of the Anteromedial Thigh

Arterial Supply

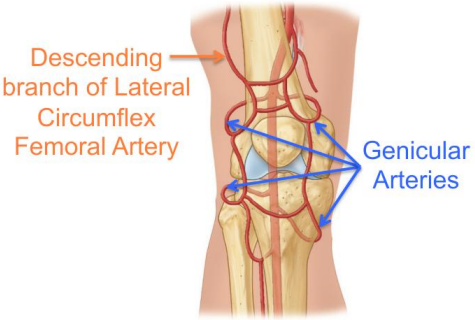


Review: Collateral Circulation

Around the Hip
Cruciate Anastomosis



Around the Knee
Geniculate Anastomosis



VI. Vasculature:

A. Femoral Artery: The external iliac artery passes deep to the inguinal ligament to become the femoral artery. Within the femoral triangle the artery gives rise to several small superficial vessels.

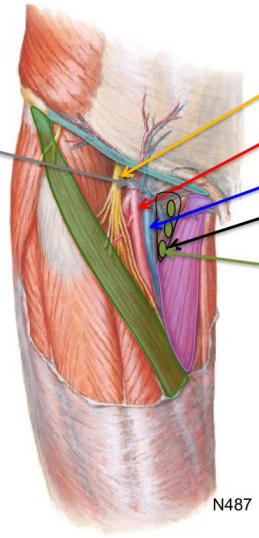
a. **Profunda Femoris Artery (Deep femoral Artery):** Major branch of the femoral artery that courses deep in the femoral triangle giving rise to the following vessels:
   i. Perforating Branches: usually 3-4 branches, arise from the deep femoral artery, that travel posteriorly to supply the posterior thigh; these branches supply all three compartments.

b. **Lateral Femoral Circumflex Artery:** mainly supplies the muscles on the lateral side of the thigh with the following three branches:
   i. ***Ascending branch*** – runs deep to rectus femoris, travels superiorly and wraps anteriorly to femoral neck to anastomose with medial femoral circumflex artery
   ii. ***Transverse branch*** – travels laterally to wind around femur
   iii. ***Descending branch*** – travels inferiorly toward the knee to anastomose with the superior lateral genicular artery

c. **Medial Femoral Circumflex Artery:** passes medially and posteriorly between iliopsoas & pectineus to reach the posterior part of the thigh and to supply most of the blood to the head and neck of the femur wraps around proximal femur to anastomose with the inferior gluteal artery.

Clinical Note: The superficial position of the femoral artery in the femoral triangle makes it vulnerable to injury. Blood supply to the lower limb can be maintained with femoral artery ligation because of the cruciate anastomosis formed by the union of the medial and lateral femoral circumflex, inferior gluteal, and first perforating arteries.

Femoral Triangle



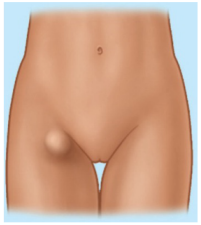
Contents:
- Femoral Nerve
- Femoral Artery
- Femoral Vein
- Femoral Canal (empty space)
- Lymphatics

Femoral Sheath
- Encloses femoral artery, femoral vein and femoral canal
- Entrance to canal = femoral ring

Relationships:
Lateral → Medial
NAVEL

Boundaries:
- Inguinal Ligament
- Sartorius
- Adductor Longus

Knowledge Check



A 52 year old woman presents to her family physician complaining of a bulge in her groin area. She notes that the bulge gets bigger when she does crunches during her exercise class. Upon physical exam you note that the bulge is inferior to her inguinal ligament.

_____

-
-
-
-

VII. Major features and landmarks

A. Femoral Ring – small, oval opening at the abdominal end of femoral triangle through which the femoral vessels emerge and enter the femoral sheath.

Clinical Note: The femoral ring is a small proximal opening of the femoral canal often filled with lymphatic vessels and fatty tissue. A loop of small intestine can enter the femoral canal through the femoral ring, causing a femoral hernia.

B. Femoral triangle – bordered by sartorius, adductor longus, and the inguinal ligament. Contents include femoral artery, femoral vein, femoral nerve, and termination of saphenous vein. Order of contents of femoral triangle from lateral to medial = **NAVEL.**

1. **N** (femoral Nerve)
2. **A** (femoral Artery)
3. **V** (femoral Vein)
4. **E** (femoral canal (Empty space))
5. **L** (Lymphatics) – within femoral canal

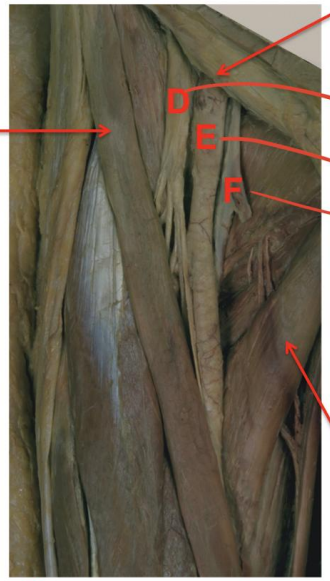
C. Femoral Sheath – continuation of the iliac fascia (overlying iliacus) and transversalis fascia ensheathing the **femoral artery, femoral vein, and femoral canal.** The femoral nerve is not found within the femoral sheath!

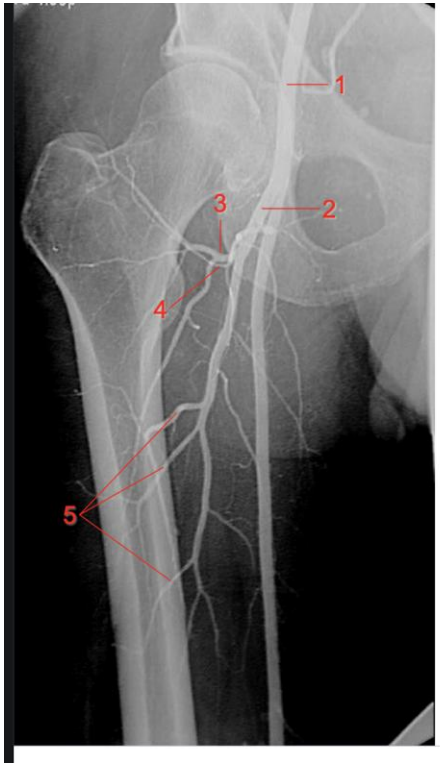
D. Adductor Canal: Formed by sartorius, vastus medialis, and adductor longus. This canal transmits the femoral artery and vein to the adductor hiatus.

E. Adductor Hiatus: opening between the distal attachment of the adductor part of adductor magnus and tendon of the hamstring part. It transmits the femoral artery and vein from the anterior compartment of the thigh to the popliteal fossa posterior to the knee. The femoral artery and vein change names to the popliteal artery and vein,

respectively.

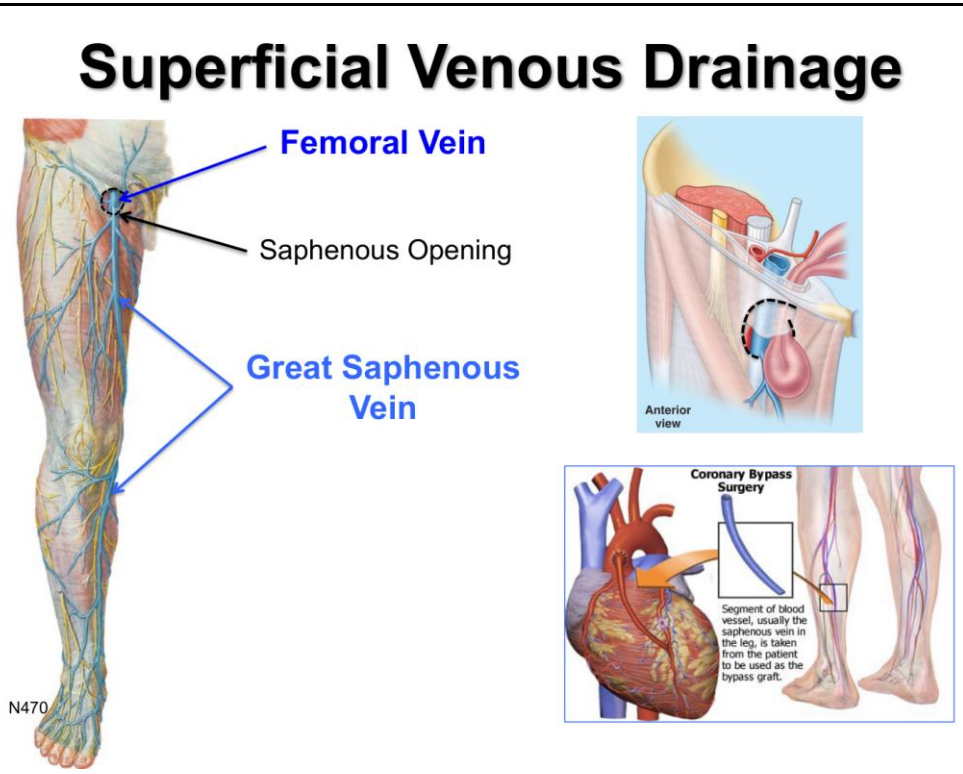
Knowledge Check





Knowledge Check

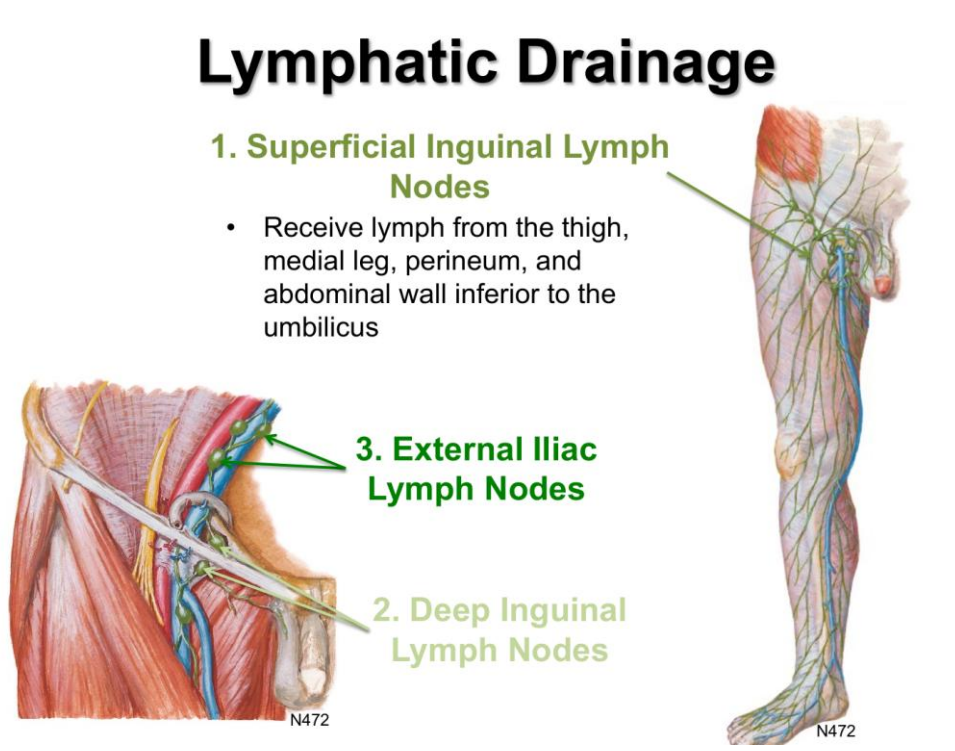
1. femoral artery — split into 2
2. profunda femoris artery
3. lateral circumflex femoral artery — 3 branch
4. medial circumflex femoral artery
5. perforating arteries



C. Femoral Vein: continuation of the popliteal vein proximal to the adductor hiatus; this deep vein ascends through the adductor canal and leaves the thigh within the femoral sheath to become the external iliac vein in the pelvis (contains valves).

D. Great saphenous vein: a superficial vein arising from the dorsal venous arch of the foot and traveling up the medial aspect of the leg and thigh alongside the ***saphenous nerve***. It ascends superficial to the fascia lata of the thigh until it passes through the saphenous opening to empty into the femoral vein (contains valves).

Clinical Note: During coronary artery bypasses, the **great saphenous vein** is often used because of its accessibility as well as the high percentage of muscular and elastic fibers that lie within its walls. During harvesting of the vein, the *saphenous nerve* is at risk of being damaged because of its close proximity to the vein.



VIII.Lymphatic Drainage

A. Superficial & deep lymphatic vessels drain lymph from the lower limb to the **inguinal lymph nodes** located in the upper portion of the femoral triangle. They are divided into two groups: 1. **Superficial inguinal lymph nodes**: located deep to fascia in the femoral triangle

a. Receive lymph from the abdominal wall below the level of the umbilicus, perineum, thigh, and medial side of the leg.
b. Most lymph from these nodes passes to the ***external iliac lymph nodes***, but some lymph may pass to the deep inguinal lymph nodes

Deep Inguinal Lymph Nodes: located on the medial side of the femoral vein. These nodes drain into ***external iliac lymph nodes***.

Take Home Points

1. The anterior compartment of the thigh functions to flex the hip and extend the knee
   a. Innervated by branches of femoral nerve
   b. Supplied by branches of the femoral artery passing through the compartment

2. The medial compartment of the thigh functions ad**d**uct the hip
   a. Innervated by branches of obturator nerve
   b. Supplied by perforating branches of the femoral artery and branches of the obturator artery

3. The hip and knee joints have extensive collateral circulation

4. Superficial venous drainage and lymphatic drainage occurs superficial to the fascia lata